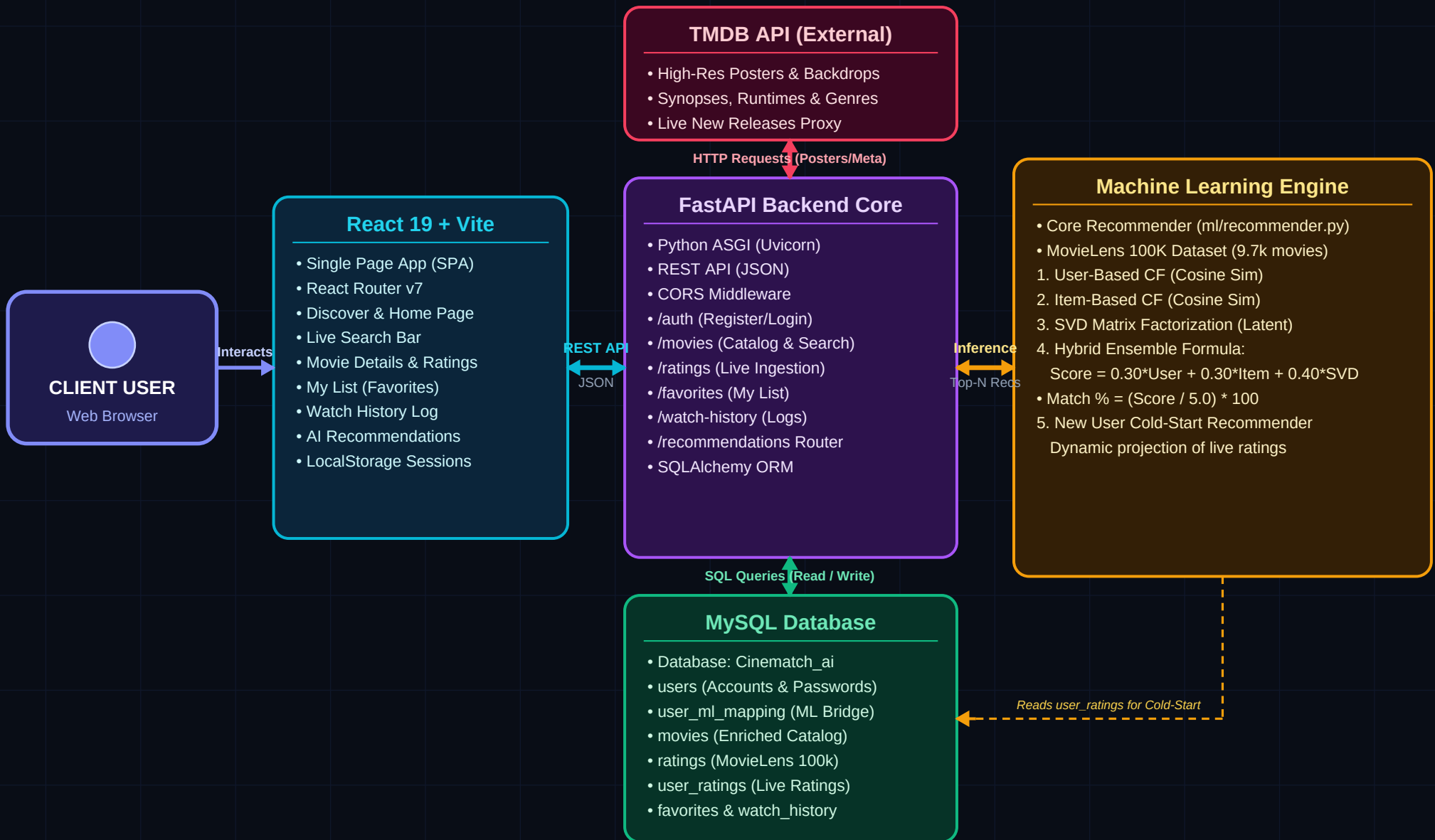


CineSuggest-AI / CineMatch-AI

End-to-End System & Machine Learning Architecture

Production Blueprint v1.0



CineSuggest-AI: Architectural Specifications & Flow

Technical reference guide documenting component responsibilities, data interfaces, and machine learning formulations for the CineMatch-AI / CineSuggest-AI platform.

Layer / Component	Technologies	Primary Responsibilities	Data Interfaces
Presentation Layer	React 19, Vite, React Router v7, Vanilla CSS	• Responsive Netflix-style UI • Discover carousels & live search view • Movie detail viewer with dynamic routing • Star ratings, watchlist & history management	Calls FastAPI REST endpoints via Native Fetch (JSON over HTTP on Port 8000)
Application Layer	FastAPI, Uvicorn, SQLAlchemy, PyMySQL, Bcrypt	• Central orchestrator for all business logic & API routes • User authentication & password hashing • Hybrid catalog search (MySQL first, TMDb fallback) • Rating, Watchlist, and History management • Recommendation dispatcher	Inbound: React HTTP Requests Outbound: PyMySQL to Database, Requests to TMDb, In-process ML calls
Machine Learning Core	Scikit-Learn, Scikit-Surprise, Pandas, NumPy, Joblib	• User-Based CF: Cosine similarity on User-Item matrix • Item-Based CF: Cosine similarity on Item-User matrix • SVD Matrix Factorization: Latent user-item factors • Hybrid Formula: $0.30 \cdot \text{User} + 0.30 \cdot \text{Item} + 0.40 \cdot \text{SVD}$ • Cold-Start Recommender: Live user rating weights	Loads MovieLens CSVs, SVD pickle model; reads live user_ratings from MySQL for users > 610
Database Layer	MySQL (Cinematch_ai)	• Persistent storage for users and application data • user_ml_mapping: Bridges app users to ML ID space • user_ratings: Live submitted user reviews • movies: Enriched metadata from TMDb	SQLAlchemy ORM session pool over localhost:3306
External Services	The Movie Database (TMDb) API v3	• High-resolution poster & backdrop image CDN • Live Now-Playing and upcoming releases • Movie synopses, runtimes, and genres	HTTPS REST API (api.themoviedb.org)

Execution Workflow & Interaction Stages

- User Registration & ML Bridge:** When a user registers (/auth/register), FastAPI creates a record in `users` and automatically maps them in `user\_ml\_mapping` with an ML ID starting from 611 (MovieLens users occupy 1-610).
- Dynamic Search Flow:** The user types in the React search bar. FastAPI checks MySQL first. If absent, it queries TMDb live, ensuring unlimited movie discovery without manual catalog entry.
- User Rating Ingestion:** When the user rates a film (1-5 stars) on the details page, the rating is committed directly to `user\_ratings` in MySQL.
- Recommendation Pipeline:** Visiting /recommendations/{user_id} inspects `ml\_user\_id`: If benchmark user (<=610), it runs the full Hybrid formula (30% User-CF + 30% Item-CF + 40% SVD). If a new app user (>610), it fetches their real ratings from MySQL, calculates cosine similarity against the MovieLens matrix, weights by rating, and normalizes scores to 0-100% Match.